

Temperature dependence of spectral induced polarization data: experimental results and membrane polarization theory

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SUMMARY

Spectral induced polarization measurements are affected by temperature variations due to a variety of temperature-dependent parameters that control the complex electrical conductivity. Most important is the influence of the ion mobility, which increases with increasing temperature. It is responsible for the increase of the conductivity of the fluid in the pores with temperature and influences the electrical double layer on the mineral surface. This work is based on laboratory measurements of 13 sandstone samples from different sources with different geological and petrophysical characteristics. We measured the complex impedance in a frequency range from 0.01 to 100 Hz and a temperature range from 0 to 40 °C. The main observation is a decrease of the characteristic time (defined by the inverse of the frequency, at which the phase shift is maximum) with increasing temperature. The strength of this decrease differs from one sample to another. The temperature dependence of the phase shift magnitude cannot easily be generalized, as it depends on the particular sample. The experimental findings suggest that neglecting the influence of temperature on complex conductivity may lead to significant errors when estimating hydraulic conductivity from relaxation time. We also simulate the temperature dependence with a theoretical model of membrane polarization and review some of the model properties, with an emphasis on the temperature dependence of the parameters. The model reproduces several features characterizing the measured data, including the temperature dependence of the characteristic times. Computed tomography and microscope images of the pore structure of three samples also allow us to associate differences in the geometrical parameters used in the modelling with pore scale parameters of the actual samples.

Key words: Electrical properties; Hydrogeophysics; Hydrothermal systems; Permeability and porosity.

1 INTRODUCTION

The complex electrical conductivity of sediments is influenced by different physical properties, including the temperature. Consequently, variations of the temperature of the subsurface may change the results of induced polarization (IP) measurements in the field. Seasonal and weather-related differences in temperature may lead to notable variations particularly for the shallow subsurface, and measurement results might not be comparable if taken at different temperatures. Relations between IP parameters and petrophysical or hydrological parameters are often based on laboratory measurements, which are usually performed at room temperature or in climate chambers. The typical temperature is around 25 °C, which does not necessarily coincide with the ambient temperature of the

subsurface under *in situ* conditions. The differences in temperature near the surface are usually small, but it is also conceivable that induced polarization measurements may be used for exploration of geothermal areas, where a large temperature range is covered, as IP is particularly suitable for estimating the hydraulic conductivity of the subsurface (e.g. Börner *et al.* 1996; Hördt *et al.* 2007).

The temperature dependence of DC conductivity was studied by several authors, using various physical and empirical approaches (e.g. Llera *et al.* 1990; Sen & Goode 1992; Hayley *et al.* 2007) and is well known. However, there is a number of temperature-dependent parameters, which influence the complex frequency-dependent electrical conductivity additionally to the DC conductivity of porous media. The complex electrical conductivity increases with increasing temperature, both in the real and imaginary part, as reported,

for example, in Binley *et al.* (2010) for temperatures between 5 and 30 °C. The time constant decreases with increasing temperature as a consequence of increasing ion mobility. This is described for timescales determined from the inverse of the spectral position of the phase maximum (Binley *et al.* 2010) and for mean relaxation times from a Debye decomposition (Zisser *et al.* 2010a) and also by Tong & Tao (2007) for IP measurements in the time domain. Tong & Tao (2007) also proposed a correction of temperature effects on the relaxation time of shaly sands by shifting the relaxation time spectra along the relaxation time axis.

Observations of measured parameters describing the polarization strength differ depending on the investigated material. Collett (1959) reported an increase of the polarization strength of andesite particles with increasing temperature for clean material, but a decrease if a small amount of pyrite is added. A similar relation was found by Vinegar & Waxman (1984) for a sandstone sample with low clay content compared to a sandstone sample with high clay content for temperatures between −5 and 95 °C. An influence of the clay content on the temperature dependence was observed also by Treichel *et al.* (2014). However, the phase shift of the sandstone samples investigated by Binley *et al.* (2010) remained constant. Worthington & Collar (1984) refer to measurements of other authors, who observed a change in the polarization with temperature that was not monotonous.

Although there are several investigations of the temperature dependence of spectral induced polarization (SIP), the existing data comprise only a few different sample materials and some of the observations differ. Especially, the behaviour of the phase shift is poorly constrained. Moreover, it is unclear if conclusions made on observations of individual samples are valid also for other materials.

The theoretical background of the temperature dependence of the complex electrical conductivity is based mainly on empirical or semi-empirical approaches, which do not cover all characteristics of the measured data. An approach to explain the temperature dependence of the phase shift can be found in Revil (2012), who describes the influence of temperature on SIP with different temperature dependencies of the ion mobilities in the free electrolyte, the diffuse layer and the Stern layer of the electrical double layer (EDL). However, a general conclusion about the temperature behaviour is not suggested.

Temperature-dependent SIP data may also be useful to test existing theories about the origin of the IP effect, including mechanistic models. Mechanistic models in turn provide a good opportunity to improve our understanding of the temperature dependence of the polarization effects in porous media. Furthermore, models can help to generalize the observations of measurements and predict the temperature dependence of SIP data.

Currently, there are principally two types of models, which describe polarization mechanisms at the pore scale; the grain-based electrochemical polarization model (e.g. Schwarz 1962; Leroy *et al.* 2008; Revil & Florsch 2010) and the pore-based membrane polarization model (e.g. Marshall & Madden 1959; Buchheim & Irmer 1979; Titov *et al.* 2002). The membrane polarization model, suggested by Marshall & Madden (1959), solves the equations describing the motion of ions for a 1-D geometry. Changing ion mobilities along the pore system leads to a concentration gradient, when an external electric field is applied. Bückner & Hördt (2013a) extended this model by explicitly parametrizing the EDL and the pore radii of cylindrical pores.

In order to gain further insight into the temperature dependence of SIP, we add additional data acquired with different materials and compare them with previous observations and empirical

relationships. We discuss the extended membrane polarization model by Bückner & Hördt (2013a) with an emphasis on the temperature dependence of the model parameters and compare measured with calculated phase spectra.

2 TEMPERATURE DEPENDENCE OF THE COMPLEX ELECTRICAL CONDUCTIVITY

There are two main factors causing the temperature dependence of the polarization response: (1) the electrical conductivity of the pore fluid, the temperature dependence of which basically stems from changes in ion mobility and (2) the EDL, coating the surface of the rock matrix or grains. The surface of the solid phase usually acquires a negative charge, when brought into contact with the pore fluid. Close to the surface within the electrolyte, the net surface charge results in an increased concentration of ions of opposite charge (counter ions). This region is known as the EDL, which consists of two parts; an inner Stern layer of strongly bound counter ions and an outer diffuse layer. Within the diffuse layer, ions can move freely but are affected by diffusion and electrostatic forces. The properties of the EDL are mainly controlled by the temperature-dependent parameters ion mobility, Zeta potential and Debye length.

For a small temperature range of 0–30 °C, the temperature dependence of the DC conductivity $\sigma(T)$ of the bulk electrolyte can be approximated by a simple linear relation (e.g. Sorensen & Glass 1987; Hayley *et al.* 2007; Binley *et al.* 2010):

$$\sigma(T) = \sigma(T_0)(1 + a(T - T_0)). \quad (1)$$

Here, $\sigma(T_0)$ is the DC conductivity (in S m^{-1}) at the reference temperature T_0 and a is a temperature compensation factor, which assumes values between 0.02 and 0.025 K^{-1} for different sample types and salinities. For a wider temperature range, the temperature dependencies of dynamic viscosity η , hydrodynamic radius and Coulomb interaction result in a non-linear relation. In this case, and for materials where surface conductivity can be neglected, the semi-empirical ‘modified Walden product’ appropriately describes the variation of the DC conductivity with temperature (e.g. Sorensen & Glass 1987).

$$\sigma(T) = \sigma(T_0) \left(\frac{\eta(T_0)}{\eta(T)} \right)^{\alpha_\sigma}. \quad (2)$$

The empirical constant α_σ is a measure of the strength of the temperature dependence. Eq. (2) is based on the assumption, that the temperature dependency of the ion mobility in the pore fluid is the basic cause of the temperature dependency of DC conductivity. The underlying relation between ion mobility and dynamic viscosity is:

$$\mu(T) = \mu(T_0) \left(\frac{\eta(T_0)}{\eta(T)} \right)^{\alpha_\mu}. \quad (3)$$

At temperatures between −20 and 110 °C and standard pressure, the dynamic viscosity of water can be calculated from the empirical equation by Pátek *et al.* (2009):

$$\eta(T) = 10^{-6} \cdot \sum_{i=1}^4 a_i \left(\frac{T}{300} \right)^{b_i}, \quad (4)$$

with parameters $a_i = [280.68, 511.45, 61.131, 0.45903]$ and $b_i = [-1.9, -7.7, -19.6, -40]$. η is given in $\text{Pa}\cdot\text{s}$ and T in K.

So far we have only considered the temperature dependence of the ion mobility and the DC conductivity of the bulk electrolyte. However, polarization effects taking place at the solid–fluid interface of a

porous medium give rise to a frequency dependence of the complex electrical conductivity. One important parameter that describes the spectral behaviour is the characteristic time τ of the polarization process, which is proportional to the square of a geometrical scale of the pores/grains as well as reciprocal to the diffusion coefficient D (e.g. Titov *et al.* 2002; Leroy *et al.* 2008; Bucker & Hördt 2013b). The relation between the diffusion coefficient and ion mobility μ is described by the Nernst–Einstein relation:

$$D = \frac{\mu k_B T}{e}. \quad (5)$$

Based on the changes in diffusion coefficient (eq. 5) and mobility (eq. 3), a first approximate relation between the characteristic time and the temperature can be derived. Following Zisser *et al.* (2010a), the characteristic time decreases with temperature as follows:

$$\tau(T) = \tau(T_0) \frac{T_0}{T} \left(\frac{\eta(T)}{\eta(T_0)} \right)^{\alpha_\tau}. \quad (6)$$

We use the symbols α_τ and α_μ for the Walden exponent if applied to the calculations of the characteristic times and mobilities, respectively. These two exponents are identical ($\alpha = \alpha_\mu = \alpha_\tau$) if the temperature dependence of the characteristic time only depends on the dynamic viscosity described by the modified Walden product. Depending on ion species and mobility, the Walden exponents range from 0.8 to 0.97 (Zisser *et al.* 2010a, and references therein).

However, eq. (6) does not account for the significant temperature variations of the Zeta potential ζ and Debye length λ_D , which control the properties of the EDL and consequently also affect the polarization process. The Zeta potential ζ is the electrical potential at the surface of shear, which is often used as the electrical potential at the interface between Stern and diffuse layers (e.g. Leroy & Revil 2004). Although the temperature dependence of the Zeta potential stems from changes in a number of parameters (e.g. pH, adsorption equilibria), the relation between Zeta potential and temperature is nearly linear (Revil *et al.* 1999):

$$\zeta(T) = \zeta(T_0)(1 + g(T - T_0)) \quad (7)$$

In the case of silica surfaces ζ increases approximately 1.7 per cent per K ($g \approx 0.017 \text{ K}^{-1}$).

The Debye length defines the characteristic spatial scale of the diffuse layer

$$\lambda_D = \sqrt{\frac{\epsilon_0 \epsilon_r k_B T}{2e c_0 z^2 F}}. \quad (8)$$

Here, $\epsilon_0 = 8.85 \times 10^{-12} \text{ As Vm}^{-1}$ is the vacuum permittivity and z the ion valence (we assume $z = 1$ throughout this work). The Debye length (eq. 8) depends on temperature both directly and indirectly, because the relative electrical permittivity ϵ_r of the pore fluid decreases with temperature (Pátek *et al.* 2009):

$$\epsilon_r(T) = \sum_{i=1}^4 e_i \left(\frac{T}{300} \right)^{f_i}. \quad (9)$$

Where $e_i = [-43.7527, 299.504, -399.364, 221.327]$ and $f_i = [-0.05, -1.47, -2.11, -2.31]$.

The effect of the temperature dependence of Debye length and Zeta potential on the complex electrical conductivity will be discussed in the context of a membrane polarization model.

3 EXPERIMENTAL DATA

Measurements were carried out on 13 different sandstone samples (see Table 1). Samples TU-B1, TU-B4 and TU-B5 stem from different drilling cores obtained at the Ketzin pilot site, Germany, (e.g. Fleury *et al.* 2013; Martens *et al.* 2013, 2014) and sample P8 is a Bunter sandstone from an outcrop in the eastern Thuringian Basin. Samples BeRo1, Biu1, FrUK1, GiUK1, koQ1, koVe1, OK1 and smVG1 are sandstones from different outcrops at the North German Basin (see Reyer & Philipp (2014) for more details). Sample S3 is an Elb Sandstone, which was previously investigated by Hördt & Milde (2012) in a different context. All samples were cut into cylinders with a diameter of 25 mm and a length of 50 mm. Available geological and petrophysical parameters of the samples are listed in Table 1. The porosities were determined from the dry and saturated weight and the volume of the samples. Dominant pore-throat diameters and the surface-area/pore-volume ratio were measured by mercury injection and nitrogen adsorption, respectively.

For three investigated rock samples (TU-B1, P8 and S3) X-ray computed tomography (CT) scans could be obtained using a GE phoenix nanotom[®] s system. The CT scans of these sandstone samples were acquired on small pieces of a few millimetres in diameter. The resolution of the scans was 1.25 μm (sample TU-B1), 1.5 μm (sample P8) and 2.0 μm (sample S3), respectively. 2-D images of the scans of the sandstone samples are shown in Fig. 1.

The main differences between the three sandstone samples are: (1) the size of the grains is smallest for sample TU-B1 and largest for S3. The grains of P8 and S3 are more rounded than the grains of TU-B1. (2) Samples P8 and S3 are more consolidated and compacted than TU-B1. The visually higher compaction is consistent with the lower porosity of sample P8 and S3 (0.11 and 0.19, for comparison: TU-B1 0.27), and the fact that P8 and S3 are robust while TU-B1 is brittle. Due to the different degrees of consolidation we assume that the pores of TU-B1 are wider than those of P8. The CT image shows that sample S3 has large pores compared to P8. This assumption is supported by mercury injection measurements, which reveal dominant pore-throat diameters of 18.3 μm for TU-B1, 2.9 μm for sample P8 and 44.4 μm for sample S3 (Table 1). However, sandstone samples TU-B1 and P8 are inhomogeneous and show regions with high cementation, where much smaller pores may exist. For estimating the pore length, we refer to the mean grain size, which is approximately 180 and 330 μm for P8 and S3, respectively (see Table 1). Because of the weak consolidation of sample TU-B1 we assume that the pores are longer than the mean grain size and similar to the length of sample P8.

3.1 SIP measurements

SIP measurements were performed with a VMP3 impedance analyser by Princeton Applied Research in the frequency range of 0.01–100 Hz. We used four-point measuring cells with non-polarizing potential electrodes as described in detail by Kruschwitz (2008), Hördt & Milde (2012) and Bairlein *et al.* (2014). The filled measuring cell was stored within a climate chamber with an accuracy of 0.1 °C to keep the temperature of the sample at the desired value. In order to assess the accuracy of the current measuring system, the measuring cell was filled with a sodium chloride solution ($\sigma_F(20^\circ\text{C}) = 50.0 \text{ mS m}^{-1}$) and measurements at temperatures between 0 and 90 °C were carried out. Fig. 2 shows that the measured electrical conductivity coincides with the increase of water conductivity with increasing temperature as predicted by the modified Walden product

Table 1. Geological and petrophysical parameters of the sandstone samples. Parameters listed are: porosity Φ , pore-fluid conductivity σ_f at 20 °C, dominant pore-throat radius r_{dom} , mean grain size d , specific surface area S_{por} , clay content, mineralogy, cementation c , sorting s and rounding r .

Sample	Local name	Φ [per cent]	σ_f [mS m ⁻¹]	r_{dom} [μm]	d [μm]	S_{por} [1 μm^{-1}]	Clay [per cent]	Mineralogy [per cent]	c [per cent]	s^c	r^c
TU-B1	Stuttgart Fm.	27.5	47.9	9.1	90	116.6	16.0 $\pm$ 10.4 ^a	36.6 $\pm$ 6.1 Q, 21.8 $\pm$ 4.6 F ^a	—	—	—
TU-B4	Exter Fm.	21.3	53.8	0.7	—	147.1	—	—	—	—	—
TU-B5	Exter Fm.	17.0	55.4	2.8	210	213.6	—	—	—	—	—
P8	Bunter Sst.	11.2	34.2	1.4	180	47.2	—	—	—	—	—
BeRo1	Rotliegend Sst.	6.9	64.6	6.0	400	64.9	3.3 ^b	59.1 Q, 4.2 F, 36.7 L ^b	17.6 ^b	4	2.5
BiSu1	Bernburg-F.	19.8	54.4	6.4	210	20.3	0.9 ^b	72.2 Q, 17.1 F, 10.7 L ^b	12.3 ^b	2	3
FrUK1	Bentheimer Sst.	14.6	57.4	5.9	100	46.6	4.3 ^b	96.1 Q, 2.1 F, 1.8 L ^b	17.0 ^b	3.5	3.5
GiUK1	Gildehaus Sst.	22.7	54.9	16.8	250	1.8	1.3 ^b	94.9 Q, 2.4 F, 2.7 L	16.8	2	2.5
koQ1	Rhät Sst.	17.1	56.9	9.7	200	20.6	0.0 ^b	78.6 Q, 4.7 F, 16.7 L ^b	31.0 ^b	1	3
koVe1	Rhät Sst.	13.0	60.6	0.9	100	96.8	0.0 ^b	83.2 Q, 3.8 F, 13.0 L ^b	40.0 ^b	1	3
OK1	Wealden Sst.	18.9	58.4	1.5	200	14.6	2.8 ^b	94.2 Q, 4.0 F, 1.8 L ^b	17.4 ^b	2	3
smVG1	Volpriehausen-F.	20.1	46.6	5.3	210	11.3	0.8 ^b	80.1 Q, 15.8 F, 4.1 L ^b	23.8 ^b	3.5	3
S3	Elb Sst.	19.3	54.8	22.2	330	9.7	—	—	—	—	—

Notes: Q: quartz, F: feldspar, L: lithoclasts.

^aFrom Förster *et al.* (2009), determined for the formation.

^bFrom Reyer (2014), measured on similar samples.

^c1: very good, 2: good, 3: intermediate, 4: poor, 5: very poor.

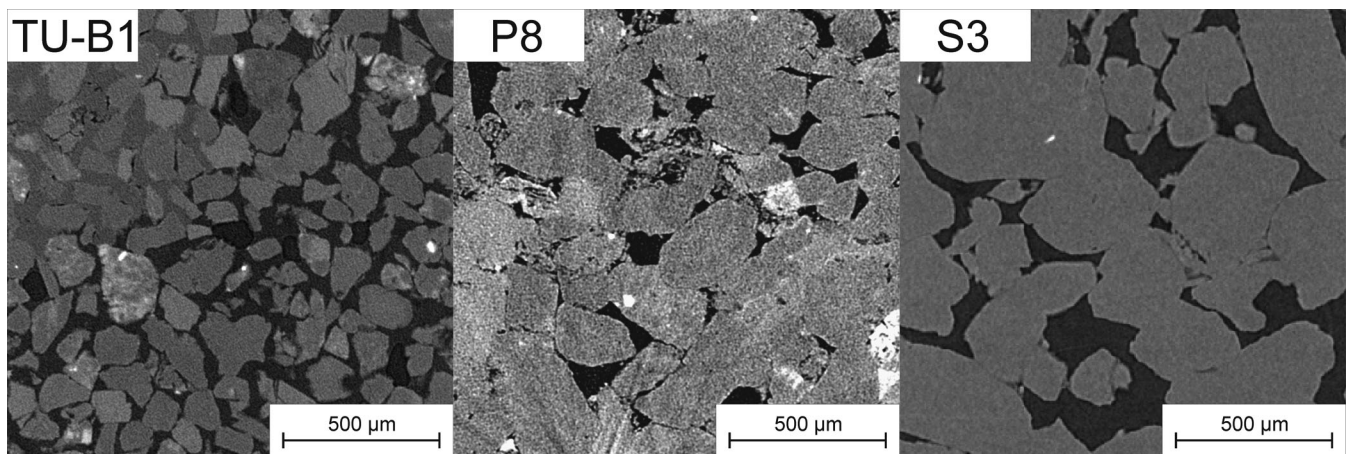


Figure 1. 2-D images of computed tomography (CT) scans of the pore space of three investigated samples. Left: sample TU-B1. Middle: sample P8. Right: sample S3.

(eq. 2) for $\alpha_\mu = 0.91$ and $T_0 = 25$ °C. This observation is independent of frequency. The measured phase shift is zero with a standard deviation of 0.2 mrad (not shown here for brevity) for frequencies between 1 mHz and 10 Hz. During the measurements, particularly at high temperatures, a small amount of water (a few millilitres) evaporated through narrow chinks between the individual parts of the sample holder, which led to a slight increase of fluid conductivity to $\sigma_f(20$ °C) = 54.9 mS m⁻¹ by the end of this test. Due to this change in fluid conductivity α_μ might be slightly overestimated, but it is still in the range between 0.8 and 0.97 given by Zisser *et al.* (2010a).

Before the actual measurements, the rock samples were dried in a vacuum chamber at 30 °C for at least 48 hr to remove residual water from the pores. Afterwards, the samples were saturated with a sodium chloride solution. The saturation process was performed slowly and under vacuum conditions in order to avoid air being trapped inside the pores. After saturation, the sample and the fluid had to equilibrate, which took several days, differing from one sample to another. When water conductivity had remained constant for at least two days, the sample was inserted into the sample holder. At the time of the measurements, the ion concentration had increased,

most probably due to the dissolution of salt that precipitated during the drying procedure and/or the dissolution of ions from the mineral matrix of the samples. Pore fluid conductivities of the samples taken before the SIP measurements are listed in Table 1.

Temperature was increased from 0 to 40 °C in steps of 5 °C. The time between increasing the temperature and starting the measurement was at least 24 hr to ensure that the sample was at the required temperature and chemical equilibrium during the measurement. Tests showed that the major change in conductivity occurred within the first two hours after a temperature change of 5 °C. Before each of these measurement series from 0 to 40 °C, one reference measurement at 20 °C was performed to assess any change of the measured spectra with time. A comparison of the reference spectra with those taken at 20 °C during the measurement series shows a slight change of electrical conductivity with time, which is supposed to arise from ions dissolved during the measurement series. The real part of the conductivity at 20 °C increases stronger during the measurements series than the imaginary part, resulting in a slight decrease of the phase shift with time.

Both real and imaginary part of the electrical conductivity increase with temperature, shown exemplarily for sample TU-B1 in

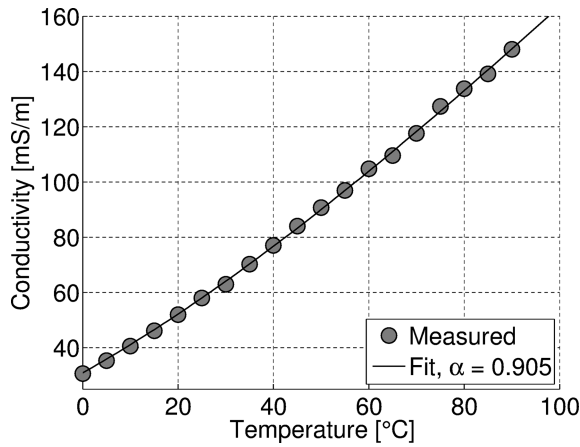


Figure 2. Magnitude of the electrical conductivity at 1 Hz versus temperature for a water-filled sample holder (circles) fitted with eq. (2) and $\alpha_\sigma = 0.905$ (solid line). The fluid conductivity at 20 °C increased during the measurement series from 50.0 to 54.9 mS m⁻¹.

Fig. 3. In agreement with eq. (1), the real part increases linearly by about 2 per cent per °C regardless of the frequency. The temperature increase of the imaginary part though depends on frequency. Its increase is largest (2.4 per cent per K for sample TU-B1) at the peak frequency f_{peak} , where the maximum of the imaginary conductivity can be observed, and smallest (1.9 per cent per K) at 0.01 Hz, where the phase shift is smallest. The different temperature dependencies of real and imaginary conductivity result in slight changes in the phase shift with temperature.

The temperature dependence of the real part of conductivity can be described quantitatively by fitting eq. (2) to the data. The Walden exponents α_σ presented in Table 2 are determined from the real conductivity at 1 Hz, which is typically used for evaluating SIP data at a single frequency (e.g. Weller *et al.* 2010). As stated before, for the real part of conductivity, the temperature dependence is independent of frequency. Most of the values are in the range previously reported in literature. Some of the samples exhibit stronger temperature dependencies ($\alpha_\sigma > 1$). Exponents > 1 mean that temperature-dependent parameters other than ion mobility become important. One possible explanation is that surface conductivity, which also depends on Zeta potential and the EDL thickness, becomes important.

The phase spectra of six of the investigated samples are displayed in Fig. 4. A common feature of the phase spectra is the increase of the frequency of the maximum phase shift with temperature, that is, the spectra shift towards higher frequencies. This is particularly obvious for all samples except GiUK1. Also the samples not presented in Fig. 3 show this behaviour. The phase spectra of sample GiUK1 appear a bit more noisy, because the phase shifts are much smaller (max. 4.5 mrad compared to up to 30 mrad for the others). Sample GiUK1 is also the only sample, which exhibits two phase maxima. While the low-frequency maximum shifts towards higher frequencies with temperature, which is in agreement with the other four samples. The high-frequency maximum appears to be independent of frequency, which might be caused by the overlay of electromagnetic coupling effects.

In order to quantify the shift towards higher frequencies and compare it with theoretical predictions, we calculated a characteristic time as the inverse of the frequency where the maximum phase shift occurs ($\tau_{\text{peak}} = (2\pi f_{\text{peak}})^{-1}$) (Fig. 5). The peak frequencies f_{peak} were determined with the help of a spline interpolation between the measured frequencies. The characteristic times decrease with

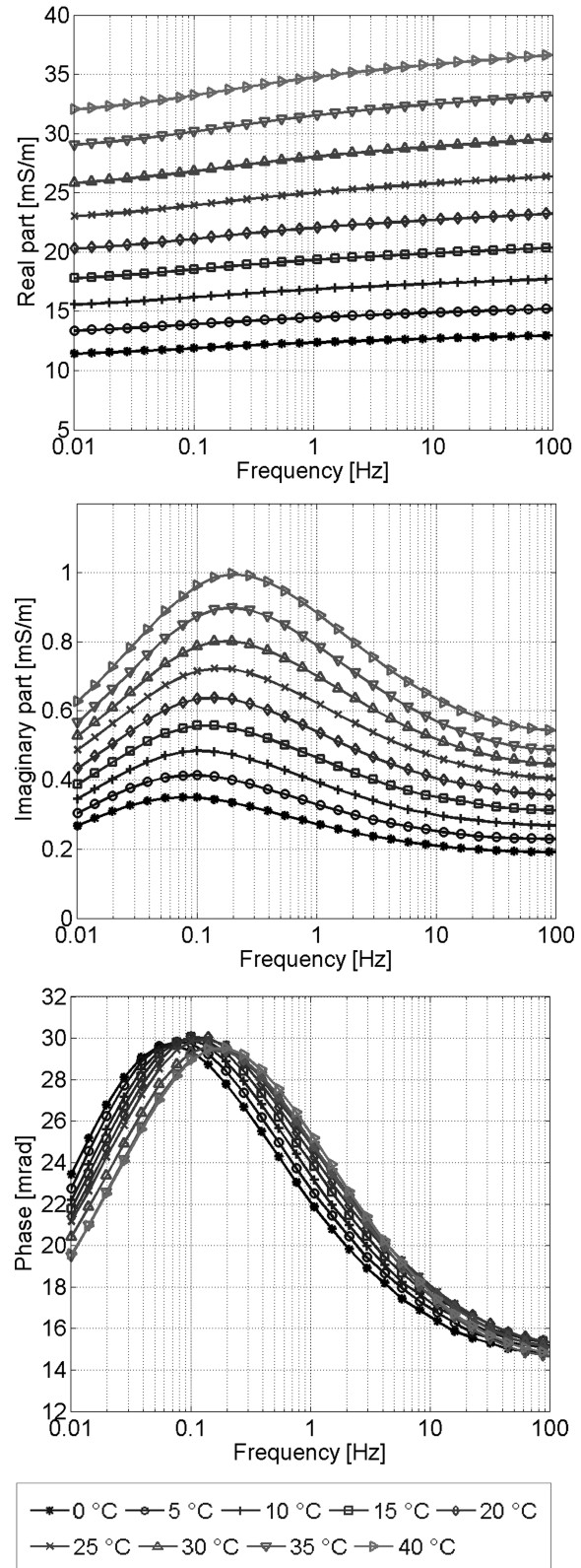


Figure 3. Real part (top), imaginary part (middle) and phase shift (bottom) of the electrical conductivity of sandstone sample TU-B1 for temperatures from 0 to 40 °C.

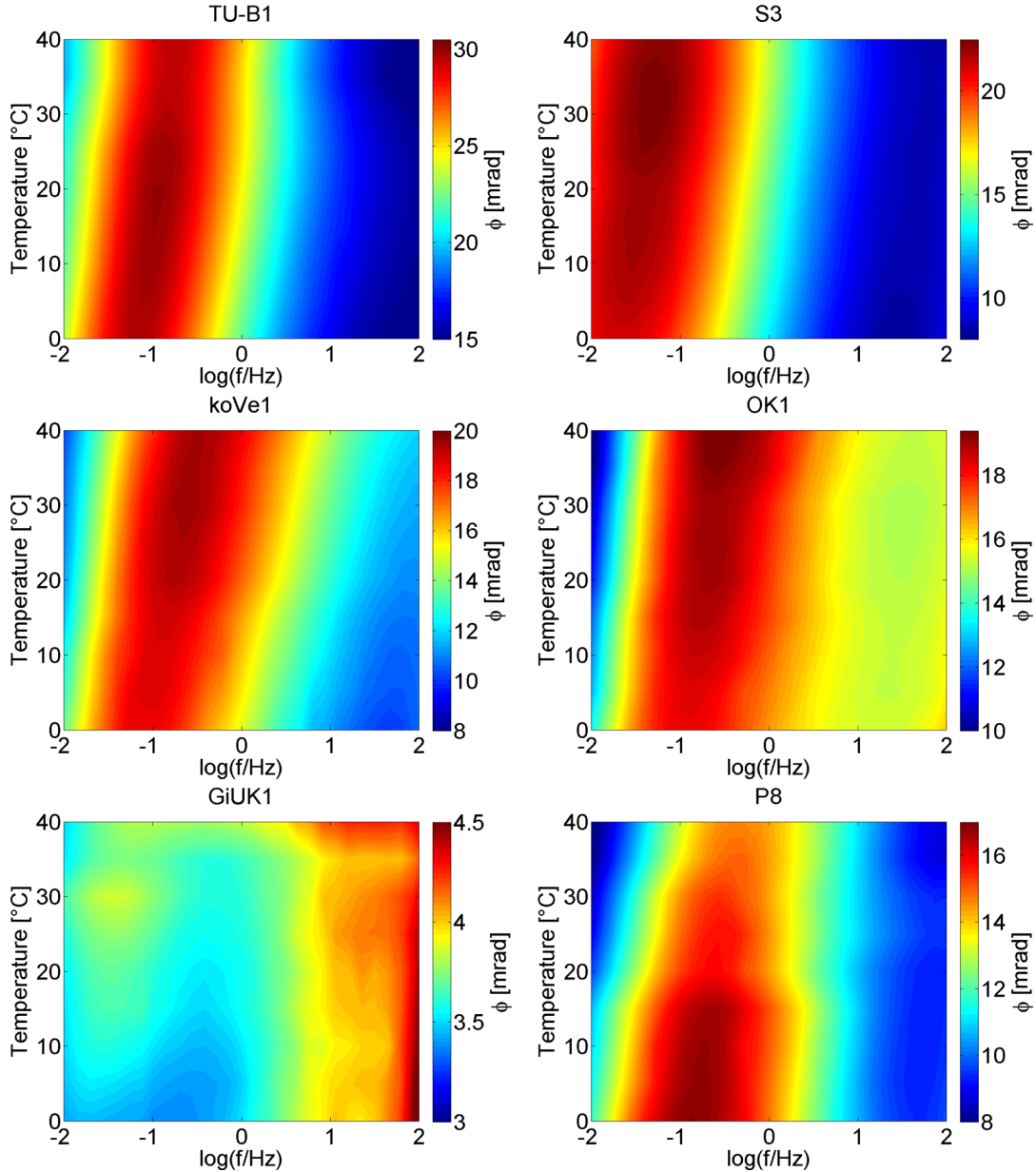


Figure 4. Phase shift of the electrical conductivity colour coded versus temperature and frequency of samples TU-B1, S3, koVe1, OK1 and GiUK and P8. Note that the colour scales are different.

increasing temperature and can be fitted with eq. (6) and Walden exponents listed in Table 2. To reduce the effect due to the error of the characteristic time at the reference temperature $\tau(T_0)$, we used every measured temperature as reference temperature to realize a separate fit. The final Walden exponents are the averages of these separate realizations along with the corresponding standard deviations $\delta\alpha_{\text{tau}}$ (Table 2). The exponents α_{tau} range from 0.52 to 1.42, which is a wider range than the values reported in literature (0.8–0.97). Compared to the real part of conductivity, the characteristic times show a wider variation in temperature dependence and also values α_{τ} that are significantly smaller than the temperature dependence of fluid conductivity ($\alpha_{\mu} = 0.905$). For some of the samples the standard deviations of the fits $\delta\alpha_{\text{tau}}$ are large, resulting from the scatter in the characteristic times, that may arise from the precision of the phase shift of 0.2 mrad.

The temperature dependencies of the amplitude of the maximum phase shifts (Fig. 4) are less uniform (i.e. the maximum phase shift can either increase or decrease with temperature) than the characteristic times. While the phase spectra of samples TU-B1, S3 and GiUK1 show an absolute maximum at 20, 35 and 30 °C, respectively, the maximum phase shift of samples koVe1 and OK1 slightly increases over the entire temperature range, indicating an absolute maximum at higher temperatures. The maximum phase shift of sample P8 decreases with increasing temperature, indicating an absolute maximum that is shifted to smaller temperatures compared to samples TU-B1, S3 and GiUK1. The temperatures of the absolute maximum in the phase shift T_{max} of all samples are listed in Table 2.

In order to analyse the cause of the differences in the temperature dependence of different samples, we examined if there is any

Table 2. Characteristic times τ_{peak} at 25 °C and Walden exponents obtained from fitting eqs (2) and (6) to the temperature dependence of the real conductivity α_σ and the characteristic times α_τ , respectively, along with standard deviations $\delta\alpha$ of the fits. T_{max} is the temperature, where the phase peak is largest within the temperature range of 0–40 °C. Sample FrUK1 does not show a maximum phase shift in the investigated frequency range.

Sample	α_σ	$\delta\alpha_\sigma$	$\tau_{\text{peak}}(25^\circ\text{C})$ [s]	α_τ	$\delta\alpha_\tau$	T_{max} [°C]
TU-B1	0.98	0.00	1.24	0.81	0.04	20
TU-B4	1.05	0.01	0.45	0.75	0.14	40
TU-B5	1.02	0.02	0.34	0.52	0.06	0
P8	1.13	0.01	0.52	0.92	0.16	0
BeRo1	1.07	0.00	1.99	1.30	0.29	15
BiSu1	0.95	0.01	0.23	0.91	0.07	10
FrUK1	0.85	0.03	—	—	—	—
GiUK1	0.87	0.01	4.12	0.92	0.15	30
koQ1	0.85	0.01	0.35	0.72	0.15	40
koVe1	0.93	0.01	0.84	1.42	0.13	40
OK1	0.97	0.00	0.71	0.83	0.08	40
smVG1	0.86	0.00	0.18	0.68	0.08	40
S3	0.88	0.02	3.83	0.89	0.06	35

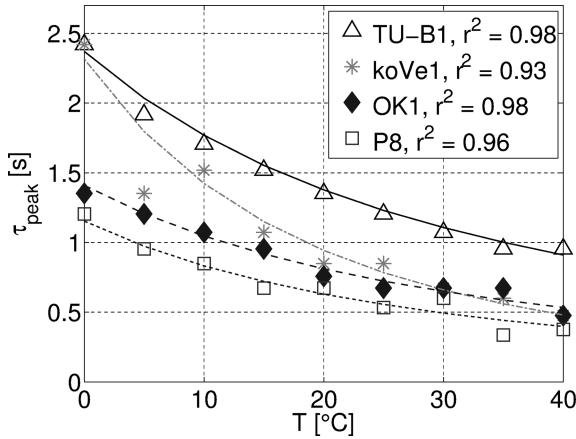


Figure 5. Characteristic times of four sandstone samples as a function of temperature. The lines show the characteristic times fitted with the modified Walden product (eq. 6), with a reference temperature of 25 °C, fitting parameters listed in Table 2 and coefficients of determination r^2 .

correlation to the sample characteristics. It seems natural that the influence of the EDL gets stronger with decreasing pore diameter, thus the Walden exponents of samples with small dominant pore-throat radius r_{dom} should deviate stronger from the Walden exponent of the electrolyte ($\alpha = 0.91$) than the exponents of samples with large pores. The temperature dependence of the real part of conductivity versus r_{dom} (Fig. 6) suggests a weak correlation of both parameters r_{dom} and α_σ , but the poor coefficient of determination ($r^2 = 0.29$) indicates that also other parameters have a strong influence. For the Walden exponents of the characteristic times, no correlation with r_{dom} could be observed. Possibly, r_{dom} is too simplified to characterize the pore space sufficiently and the entire pore- (or grain) size distribution needs to be taken into account. For none of the remaining parameters in Table 1, we could find a direct correlation to the temperature dependence of α_σ or α_τ . However, we cannot exclude that a combination of surface characteristics and the geometry of the pore space determines the temperature dependence.

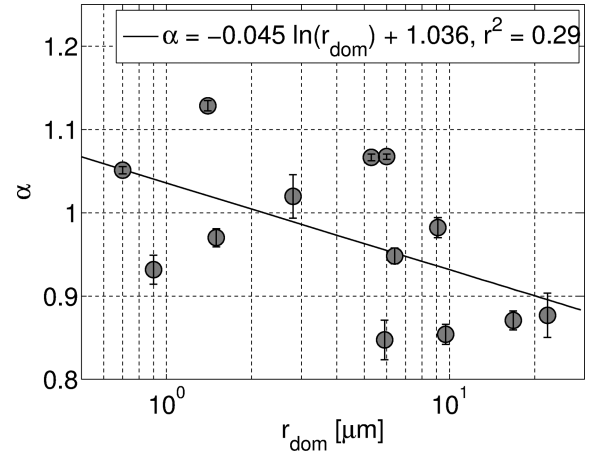


Figure 6. Walden exponent of the real conductivity versus dominant pore-throat diameter. The solid line corresponds to the logarithmic fit.

4 MODELLING THE TEMPERATURE DEPENDENCE

Our experimental results suggest that the temperature dependence of SIP data cannot be explained solely by the temperature dependence of ion mobilities. In order to obtain a deeper theoretical understanding, we carry out simulations with an existing membrane polarization model that allows to simulate temperature dependence, including the EDL properties. While we do not expect to obtain a quantitative fit to the data, we hope to be able to explain at least some of the observations with processes at the pore scale.

4.1 Membrane polarization model

The theory of membrane polarization, first established by Marshall & Madden (1959), is based on a model of alternating active and passive zones. It considers electromigration and diffusion processes within a symmetric electrolyte. Varying ion mobilities are used to describe the ion-selective (or membrane-like) behaviour of the active zones. When subject to an external electric field, an ion concentration gradient builds up resulting in a frequency-dependent impedance of the 1-D pore model. Using the approach of Bückner & Hördt (2013b), the Marshall–Madden impedance Z_{MM} can be written in terms of its DC limit R_0 (in Ωm^2), a dimensionless polarizability η_0 and two time constants τ_i (in s).

$$Z_{\text{MM}}(\omega) = R_0 \left[1 - \eta_0 \left(1 - \frac{\frac{L_1}{\tau_1} + \frac{L_2}{\tau_2}}{\frac{L_1}{\tau_1} \sqrt{j\omega\tau_1} \coth \sqrt{j\omega\tau_1} + \frac{L_2}{\tau_2} \sqrt{j\omega\tau_2} \coth \sqrt{j\omega\tau_2}} \right) \right] \quad (10)$$

$$R_0 = \frac{k_B T}{ec_0 F} \left[\frac{L_1}{D_{p1} + D_{n1}} + \frac{L_2}{D_{p2} + D_{n2}} + \frac{8(t_{n2}t_{p1} - t_{n1}t_{p2})^2}{\frac{L_1}{\tau_1} + \frac{L_2}{\tau_2}} \right], \quad (11)$$

$$\eta_0 = \frac{k_B T}{ec_0 F} \frac{8(t_{n2}t_{p1} - t_{n1}t_{p2})^2}{\frac{L_1}{\tau_1} + \frac{L_2}{\tau_2}} \frac{1}{R_0}. \quad (12)$$

$$\tau_i = \frac{L_i^2}{8D_{pi}t_{ni}} \quad (13)$$

Here, ω is the angular frequency of the external electric field (in rad s⁻¹), $k_B = 1.38 \times 10^{-23}$ J K⁻¹ the Boltzmann constant, T the absolute temperature (in K), $e = 1.6 \times 10^{-19}$ C the elementary charge, c_0 the bulk concentration in the free electrolyte (in mol m⁻³), $F = 96,500$ C mol⁻¹ the Faraday constant and $j^2 = -1$. L_1 and L_2 are the lengths (in m) of the two zones (the subscripts 1 and 2 denote the properties of the passive and the active zone, respectively). The impedance is calculated in units of Ωm^2 , because it is related to a 1-D pore system and scaled to the cross-sectional area of the pores. Due to the varying ion mobilities, the dimensionless transference numbers t_{pi} and t_{ni} , which describe the contribution of the cations and the anions to the total current, also differ between active and passive zone.

The original model by Marshall & Madden (1959) was designed to roughly estimate the magnitude of possible membrane polarization processes. The authors postulated the required mobility variations along the pores without suggesting a mechanism or quantitative estimates. Therefore, Bückner & Hördt (2013a) explicitly attributed the ion-selective behaviour of the active zone to the EDL, which coats the solid–fluid interface of porous media. In the following, we briefly review their model with an emphasis on the temperature dependence. The model describes the pore space as a sequence of wide and narrow cylindrical pores and allows to include pore radii and the properties of the EDL explicitly into the original membrane polarization model.

The electrical potential φ (in V) within the diffuse layer coating the inner surface of a cylindrical pore with radius r_1 (in m) is given by (Hunter 1981)

$$\varphi(r) = \zeta \frac{J_0(j\kappa r)}{J_0(j\kappa r_1)}, \quad (14)$$

where J_0 is the zero-order Bessel function. The decay constant κ is the inverse of the Debye length (eq. 8). The variations of the ion concentrations over the pore cross-section are then calculated from (e.g. Butt *et al.* 2003).

$$c_{n,p}(r) = c_0 \exp\left(\frac{\pm e\varphi(r)}{k_B T}\right) \quad (15)$$

In narrow pores with a radius in the order of some tens of Debye lengths or smaller, the excess of cations (in the case of a negative surface charge) and the lack of anions within the diffuse layer will result in a significant difference between average cation and anion concentrations. For a cylindrical pore with radius r_i , the average ion concentrations, normalized by the bulk concentration c_0 and the cross-sectional area of the wide pore A_1 (in m²), write

$$\bar{b}_{n,p} = \frac{1}{c_0 A_1} \int_0^{r_i} c_{p,n}(r) 2\pi r dr. \quad (16)$$

Although cations within the Stern layer are strongly bound, they can move tangentially to the surface and therefore contribute to the total cation-carried current along the pore. In order to consider this contribution, we use the partition coefficient f_Q defined by Leroy *et al.* (2008) as the ratio of charge density in the Stern layer and total charge density in the EDL (Stern and diffuse layers). The normalized average cation concentration then becomes

$$\tilde{b}_p = \frac{\bar{b}_p - f_Q}{1 - f_Q}. \quad (17)$$

As Bückner & Hördt (2013a) show, the normalized average concentrations can be used to define average transference numbers

$$t_p = \frac{\mu_p \tilde{b}_p}{\mu_p \tilde{b}_p + \mu_n \bar{b}_n} \quad (18)$$

$$t_n = \frac{\mu_n \bar{b}_n}{\mu_p \tilde{b}_p + \mu_n \bar{b}_n} \quad (19)$$

and average diffusion coefficients

$$D_p = \frac{\mu_p \tilde{b}_p k_B T}{e} \quad (20)$$

$$D_n = \frac{\mu_n \bar{b}_n k_B T}{e} \quad (21)$$

for each of the two cylindrical pores. Bückner & Hördt (2013a) showed that using the above defined average transference numbers and diffusion coefficients the analytical solution (eqs 10–13) can be used to describe the impedance of a sequence of two cylindrical pores with different pore radii.

So far, the impedance magnitude Z_{MM} was given in Ωm , because it is based on a 1-D pore system. Here, we suggest to convert it into an effective conductivity that enables us to compare the conductivity magnitudes to measured data. The conversion requires an assumption about the porosity Φ . If L_1 and L_2 are the pore lengths and A_1 and A_2 the cross-sectional areas we show in Appendix A that the effective conductivity may be defined by:

$$\sigma_{\text{eff}} = \frac{A_1}{Z_{MM}} \cdot \frac{L^2 \Phi}{A_1 L_1 + A_2 L_2}. \quad (22)$$

The temperature dependence of the model parameters was included by implementing the temperature dependence of mobility, Zeta potential, Debye length and permittivity as described in chapter 2.

4.2 Modelling results

In order to explore how the overall polarization process responds to the temperature dependencies of the different model parameters described above, we carry out a modelling study. If no other value is specified, the following model parameters will be used: pore lengths are chosen to fit the typical pore size of a fine-grained sandstone (e.g. Howard *et al.* 1993; Lindquist *et al.* 2000; Peng *et al.* 2012) with the length of the wide pore $L_1 = 50$ μm and the length of the narrow pore $L_2 = 10$ μm . Pore radii are chosen to cause a significant phase response of the model impedance ($r_1 = 0.2$ μm and $r_2 = 0.02$ μm for the wide pore and the narrow pore, respectively). The ion concentration of the pore fluid was set to 1 mol m⁻³. An ion mobility of $5 \times 10^{-8} \frac{\text{m}^2}{\text{Vs}}$ was used, which is approximately the mobility of a Na⁺ ion (e.g. Atkins & De Paula 2013). Following Leroy *et al.* (2008), a Zeta potential of -75 mV and a (constant) partition coefficient of 0.6 were used. For calculating the ion mobility (eq. 3), we used a Walden exponent of $\alpha_\mu = 0.91$, which was the best matching value for the data of Zisser *et al.* (2010a). The reference temperature T_0 for the calculation of the temperature dependence of the mobilities and the Zeta potential (eq. 7) was 25 °C. The porosity for calculating the effective conductivity was $\Phi = 0.2$, which is a typical value for sandstones.

For all frequencies in the range from 10 mHz to 100 Hz, the real part of the effective conductivity increases monotonously with temperature (Fig. 7). The modeled temperature dependence of the real part of the effective conductivity can be fitted by a Walden product (eq. 2) with an exponent $\alpha_\sigma \approx 2.19$, which is large compared to the value that was chosen for the ion mobility in the free electrolyte (0.91). This large value implies, that the temperature dependence of the effective conductivity is strongly influenced by surface conductivity and cannot be described solely by the ion mobility in the

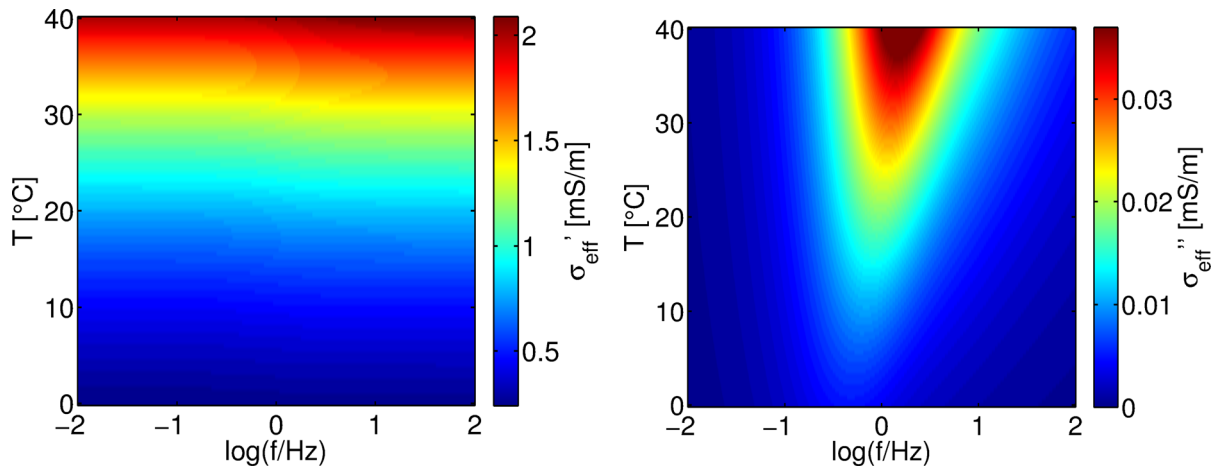


Figure 7. Effective conductivity of the model colour coded versus temperature and frequency calculated with parameters $L_1 = 50$, $L_2 = 10$, $r_1 = 0.2$ and $r_2 = 0.02 \mu\text{m}$. Left: real conductivity. Right: imaginary conductivity.

free electrolyte. The temperature dependence of the surface conductivity is stronger than the dependence of the fluid conductivity on temperature. We conclude that besides the mobility other factors, most likely the Zeta potential and Debye length, also control the admittance. The conclusion is supported by the observation that the temperature dependence of the modeled real conductivity adopts the value of the mobility (0.91) for large pore radii ($r_1, r_2 > 100 \mu\text{m}$, not shown here for brevity) where the double layer becomes unimportant. The imaginary part of the effective conductivity also increases with temperature, depending on the frequency.

Fig. 8 shows the variation of the phase shift with frequency and temperature for three different pore lengths. The ratio of the two pore lengths L_1 and L_2 is kept constant, because it is more likely that in natural materials with long pores both narrow and wide pores are long compared to materials with short pores. The maximum phase shift attains highest values at medium temperatures (approximately 30°C), which remain unchanged when the pore length is increased. Moreover, for all studied pore lengths, the peak frequency f_{peak} , where the maximum phase shift occurs, increases with increasing temperature. This translates into a decrease of the characteristic time (Fig. 9) with increasing temperature (calculated the same way as for the measured data), resulting mainly from the increase of ion mobility and diffusion coefficient as described by eq. (6). As reported before by B  cker & H  rdt (2013a,b), the critical frequency of the membrane polarization model shifts towards lower frequencies, when increasing the pore length. Fitting the characteristic times with eq. (6) results in a slightly stronger temperature dependence of $\alpha_\tau = 1.07$ than $\alpha_\mu = 0.91$ used for the ion mobility. As for the admittance, mobility is not the only factor controlling the characteristic times.

Fig. 9 also shows that it is possible to fit the behaviour of the characteristic time of the data (using P8 as an example) almost exactly by choosing appropriate pore lengths. Measured and simulated curves of the characteristic times can be fitted with the modified Walden product with exponents of 0.92 (sample P8) and 1.07 (model).

Fig. 10 shows the variation of the phase shift with frequency and temperature for three different pore radii. Note that the ratio between the pore radii of the two pores is kept constant. Again, the peak frequency increases with temperature. Changing the pore radii in the range presented here has no influence on the characteristic timescale and its temperature dependence (not shown here). The absolute phase maximum though moves to higher temperatures when increasing pore radii, and changes its magnitude.

5 DISCUSSION

Our measurement results exhibit a clear dependence of SIP parameters on temperature. The real part of the complex electrical conductivity of our measured IP spectra shows the expected increase of approximately 2 per cent per $^\circ\text{C}$, which is in agreement with previous observations of other authors (e.g. Hayley *et al.* 2007; Binley *et al.* 2010). Similarly, the imaginary part increases about 2 per cent in average (consistent with Binley *et al.* 2010), but the strength of the increase depends on frequency. The small deviations of the temperature dependencies of the imaginary part from the real part leads to a slight change in the magnitude of the phase shift with temperature.

For the changes in chargeability with temperature, Worthington & Collar (1984) reported both an increase and a non-monotonic behaviour for different studies. As the phase shift is comparable to chargeability, obtained from time-domain measurements, these results are consistent with the increase of the maximum phase shift and the absolute maximum found here for some of the samples (Fig. 4). The temperature dependence of the magnitude of the phase shift cannot be generalized. Our data show that both increasing and decreasing phase shifts with increasing temperature can occur, which was also observed by Vinegar & Waxman (1984) on two samples with different clay content. However, for the samples investigated here, we could not find a correlation of the temperature behaviour of the phase shift and the clay content. Binley *et al.* (2010) reported a temperature-independent phase shift (magnitude) for a small temperature range. This is not a contradiction to our results, as the change in the phase shift in our data is weak and could be interpreted as irrelevant for small temperature changes of a few degree centigrade. Nevertheless, for large temperature differences, a significant change in the phase shift of several milliradians can occur, which may lead to misinterpretations of measurement data if neglected. Additionally, a detailed observation of the temperature dependence of the phase shift helps to understand the temperature dependence of the underlying polarization process.

5.1 Characteristic time of measured spectra

From the CT images of sandstone samples TU-B1, P8 and S3, we can make some qualitative statements about the average pore size of the samples relative to each other. The characteristic times are associated with a characteristic geometrical scale of the pores or

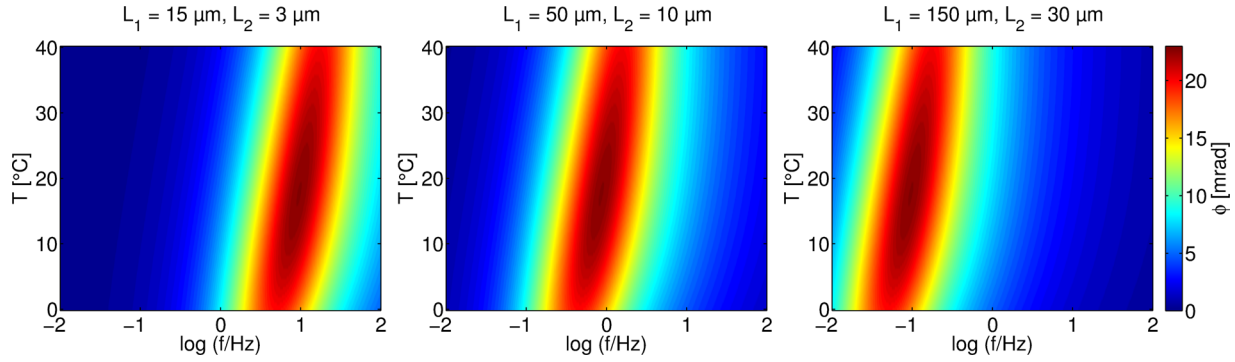


Figure 8. Phase shift of the modeled admittance colour coded versus temperature and frequency for different pore lengths L_1 and L_2 . Pore lengths increase from left to right. Pore radii are $r_1 = 0.2$ and $r_2 = 0.02$ μm .

grains. For the membrane polarization model, the dominating factor is the length scale of the pores. Pores of sandstone samples TU-B1 and P8 are estimated to have a similar average length, while S3 has longer pores (Fig. 1). Consequently, the maxima of the phase shift of samples TU-B1 and P8 should occur at similar frequencies and for sample S3 at smaller frequencies. At 25 °C, the peak frequency of sample P8 is found at 0.31 Hz and the one of TU-B1 at 0.13 Hz, which translates into characteristic times of 0.51 and 1.20 s (Table 2), respectively. Even though the mean grain size of sample TU-B1 is smaller than the one of sample P8, the latter is much more compacted and cemented, which seems to result in shorter pores and explain the shorter characteristic timescale. The characteristic time at 25 °C of sample S3 is larger than τ_{peak} of the two other samples and thus consistent with the geometrical considerations.

Previous publications (e.g. Scott & Barker 2005; Kruschwitz *et al.* 2010; Revil 2013) related the time constant to the dominant pore-throat diameter $D_{\text{dom}} = 2r_{\text{dom}}$. Fig. 11 shows that our data are consistent with their results, as we also see an increase of the characteristic time τ_{peak} with increasing D_{dom} . A fit of the same form as that of Scott & Barker (2005) can be obtained. This observation does not contradict the membrane polarization model, where the characteristic time is dominated by the pore length. Pore lengths are experimentally not accessible, except indirectly, because pore throats and pore lengths in real rocks usually are not independent from each other. Therefore the correlations between τ_{peak} and D_{dom} do not allow the conclusion that D_{dom} , and not the pore length, is in fact the controlling parameter at the pore scale. For sandstones with similar shaped grains and cementation, smaller pore radii are associated with shorter pores. Moreover, the fitting parameters differ from Scott & Barker (2005) and the scattering of the data are indications for the influence of other or additional parameters than D_{dom} on the time constant.

For increasing temperature, a decrease of the characteristic times was found for all samples that showed a peak in the measured phase spectra, agreeing with the results of Binley *et al.* (2010) and Zisser *et al.* (2010a). However, the strength of the temperature dependence varies for different samples. In order to quantify the temperature dependencies of the different characteristic times, they were fitted using eq. (6). The exponents α_τ range from 0.52 to 1.42, which clearly exceeds the range from 0.8 to 0.97 reported in Zisser *et al.* (2010a). The equation that was used for fitting the temperature dependence of the characteristic times (eq. 6) is based on the assumption that the only temperature-dependent parameter is the ion mobility. Zisser *et al.* (2010a) assumed, that the ion mobility in the Stern and diffuse layers are similar to the mobility in the free electrolyte, valid for clay-free sandstones. Our results with Walden

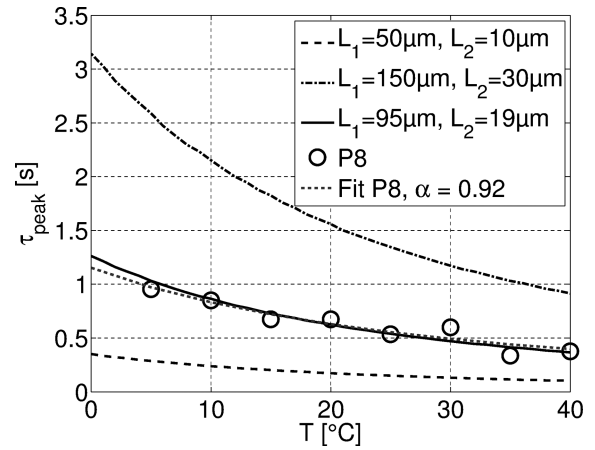


Figure 9. Modeled characteristic times of three different combinations of pore lengths L_1 and L_2 compared to the characteristic times of sample P8, fitted with $\alpha_\tau = 0.92$. The pore radii are $r_1 = 0.2$ and $r_2 = 0.02$ μm .

exponents different from that of the free electrolyte ($\alpha_\mu = 0.905$), suggest that this assumption is not valid for most of the samples of this study. However, we could not find a direct relation to the clay content or other petrophysical parameters.

5.2 Comparison of theoretical and experimental results

Although we are able to fit the temperature dependence of the characteristic time of the model to that of a sandstone sample with realistic pore lengths (Fig. 9), the following considerations are not intended to demonstrate that the model exactly describes all mechanisms taking place in the pores of a sandstone. The model represents a strong simplification of the pore geometry of a sandstone and might therefore underestimate or neglect the influence of several contributing factors. However, a qualitative comparison of the temperature dependence of the IP spectra with the model can help to assess whether the theory describes relevant processes, and possibly help to understand the processes at the pore scale underlying the observed temperature dependencies.

The effective conductivity of the model (Fig. 7) is about one order of magnitude smaller than the measured conductivity (Fig. 3). This discrepancy is small considering the simple model geometry compared to the complex network of different-sized pores in series and parallel of a natural rock. Comparing the temperature dependence of the real part of the measured conductivity and the effective model conductivity, respectively, gives a qualitative agreement in

the increasing behaviour for both quantities, but a strong difference of the Walden exponent with $\alpha_\sigma(\text{measurements}) = 0.85\text{--}1.13$ and $\alpha_\sigma(\text{model}) = 2.19$. Although the Walden exponent of the model is only valid for one pore size and can change when the pore size is changed, exponents similar to the ones measured only occur when both pore radii are larger than $1\text{ }\mu\text{m}$. At large pore radii ($r_i \gg \lambda_D$), the EDL covers a negligible part of the pore cross section. Therefore, bulk conductivity controls the absolute value of the admittance and the temperature dependence only stems from ion mobilities and diffusion coefficients (Walden exponent $\alpha_\mu = 0.91$). At much smaller pore radii ($\ll 0.1\text{ }\mu\text{m}$), as used in this study to produce sufficiently large phase shifts, the EDL covers a considerable part of the pore cross-section and the temperature dependence of the total current through the pore system also depends on the properties of the EDL. In particular, the Zeta potential increases rapidly with temperature and leads to a much stronger temperature dependence ($\alpha_\sigma = 2.19$) of the absolute value of the admittance than predicted by the modified Walden product. The difference in the Walden exponents between theoretical and measured values might imply that the model in its present form tends to overestimate the influence of narrow pores.

In the following we focus on the phase shift. The most prominent similarities of the measurements and the model are (1) the shift of the phase maximum to higher frequencies (which is equivalent to a decrease of the characteristic time) with increasing temperature and (2) different positions of the absolute maximum on the frequency and temperature axis for different samples (measurements) or pore geometries (model). These two aspects are discussed in more detail below.

5.2.1 Characteristic time

A detailed discussion of the increase of the modeled characteristic time (decrease of the peak frequency) with the pore length can be found in Bückner & Hördt (2013b). As the polarization process is mainly controlled by diffusion, the characteristic time is proportional to the square of the characteristic length of the system; $\tau \propto L^2$, with L being the total pore length.

Figs 5 and 9 show that both the characteristic times of the model and those of the sandstone samples decrease with temperature. The fitted exponent (eq. 6) of $\alpha_\tau = 1.07$ for the characteristic times of the model lies within the range of $0.52\text{--}1.42$ of the exponents fitted to the characteristic times of the sandstone samples. It is slightly higher than the Walden exponent used for the calculation of the temperature dependence of the ion mobility ($\alpha_\mu = 0.91$). This means, that also for the characteristic time, the temperature dependence of the model is not only controlled by ion mobility and diffusion coefficient, but also by the properties of the EDL.

5.2.2 Influence of pore geometry

The membrane polarization model predicts a strong temperature dependence of the magnitude of the maximum phase shift, which is related to polarizability. From eq. (12), it is evident that the polarizability (and thus the maximum phase shift) depends on a complex combination of the geometrical model parameters (pore lengths and radii) and the properties of the EDL. Earlier studies (Bückner & Hördt 2013b) revealed that the polarization magnitude adopts a maximum for certain combinations of pore radii r_1 and r_2 depending on the properties of the EDL. Our modelling results show that the temperature, at which the phase shift reaches a maximum, strongly depends on pore radii (Fig. 10). Apparently, at the temperature of

maximum phase shift, the combination of Zeta potential and Debye length maximizes the polarizability for a certain geometry (predominantly pore radii). When the pore radii increase, the fraction of the EDL in the cross-section of the pore becomes smaller and the influence of the EDL decreases. Presumably, this decrease is compensated by a higher magnitude of the Zeta potential, that is reached at a higher temperature, shifting the absolute maximum to a higher temperature.

The measurements of samples TU-B1, P8 and S3 are also consistent with our modelling results, when taking into account the pore structure as observed from the CT images. The measured phase shifts of samples TU-B1 and S3 show an absolute maximum at approximately 20 and 35°C , respectively. The phase shift of sample P8 has its maximum at a lower temperature compared to TU-B1 and S3 (Fig. 4). Actually, pores and pore throats of sample TU-B1 are narrower than those of sample S3 and wider than those of sample P8, which is in agreement with the prediction of the model. However, this observation does not hold for some of the samples, when only the dominant pore-throat radius is considered, for example, sample TU-B4 with small r_{dom} but a maximum at 40°C .

Another observation when comparing simulated and measured spectra is the different width of the phase peak, which arises from the pore-size distribution. The model considers only two pore lengths (and two radii), which result in a superposition of two characteristic times and corresponding phase peaks. However, Bückner & Hördt (2013b) point out, that generally only one of the pores dominates the relaxation process, which gives rise to a very narrow phase peak with steep slopes. In natural rock samples, on the other hand, a wide range of pore lengths may exist resulting in a distribution of characteristic times and a broadened phase peak. Similarly, the wide range of different pore radii in natural rock samples is supposed to cause a broad maximum of the phase shift in the temperature domain, leading to a weaker temperature dependence of the measured phase shift compared to the modeled one.

5.2.3 Properties of the model

The parameter sets used here to produce results comparable to the measured data include ratios between pore lengths and pore radii of up to 500. This seems relatively large, considering that typical pore-throat sizes of sandstone determined by mercury injection measurements are only about $1/10$ of the mean grain size (e.g. Binley et al. 2005; Nelson 2009). The mean grain size, in turn, is a suitable approximation for the mean pore length. The ratio $1/10$ is also in agreement with the grain and pore sizes observed in the CT images of our sandstone samples.

The CT images give an idea about characteristic length scales, but they do not include information on the possible existence of submicron pores due to the resolution of approximately $1\text{ }\mu\text{m}$. The pore radii required for calculating a phase shift similar to measured phase shifts are smaller than mean pore radii of rock samples. This discrepancy could be attributed to an underestimation of polarization effects in large pores by the model, but it is also possible that only very small pores are responsible for the polarization effect. The existence of submicron pores in our samples has been observed by the mercury injection measurements, although in most samples the dominant pore-throat diameter is larger. Also other authors have reported submicron pores for sandstone samples (e.g. Bloomfield et al. 2001; Kruschwitz et al. 2010). Anyway, at this stage we cannot say whether these submicron pores are relevant to the overall polarization response of our samples or not.

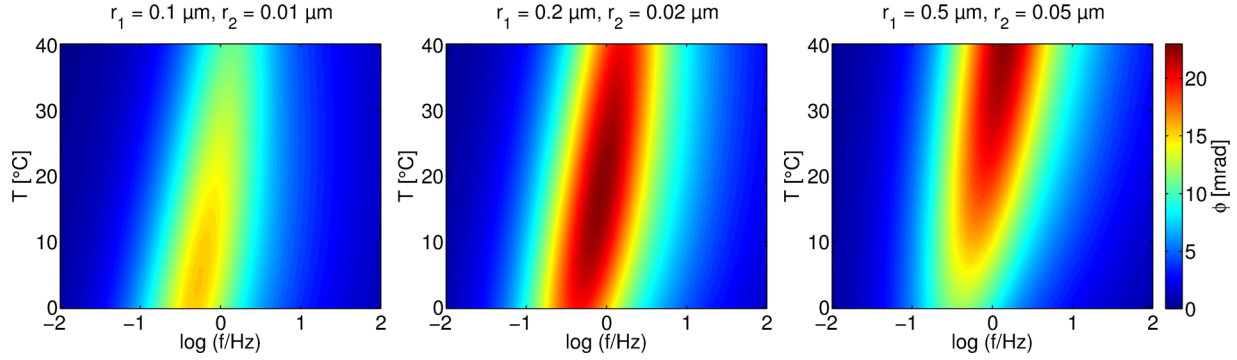


Figure 10. Phase shift of the modeled admittance colour coded versus temperature and frequency for different pore radii r_1 and r_2 . Pore radii increase from left to right. Pore lengths are $L_1 = 50$ and $L_2 = 10$ μm .

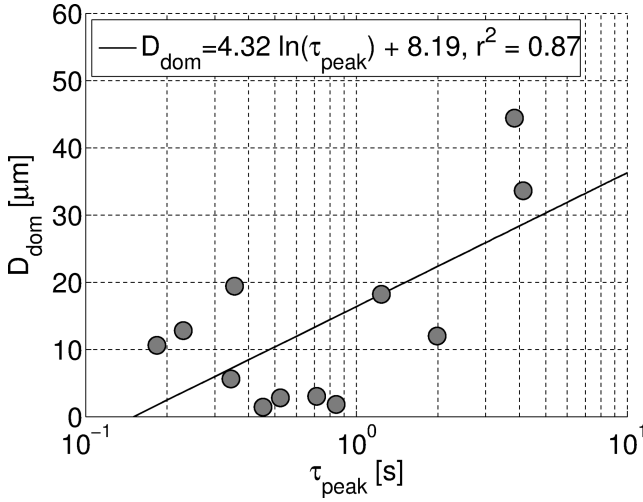


Figure 11. Dominant pore-throat diameters from mercury injection versus characteristic times of the IP spectra with logarithmic fit.

The model response is calculated with the assumption that the mobility in the Stern layer is equal to the mobility of the ions in the free electrolyte. This assumption is used for clean sands in several publications (e.g. Leroy *et al.* 2008; Revil & Florsch 2010), but it might be different in the presence of clays. For example, Revil (2013) calculate a considerably smaller ion mobility of $\mu_s = 1.5 \times 10^{-10} \frac{\text{m}^2}{\text{Vs}}$ for the Stern layer of clayey sandstones. In the membrane polarization model, an ion mobility in the Stern layer much lower than in the diffuse layer leads to $\tilde{b}_p \approx \bar{b}_p$ (see Appendix B). This means that the Stern layer does not contribute significantly to the overall conduction, if the mobility in the Stern layer is much lower than in the bulk electrolyte and the diffuse layer, which contradicts our previous idea that surface conductivity might play an important role for the temperature dependence for samples with clay content.

We conclude that the interplay between Zeta potential, partition coefficient and Stern layer mobility may not be fully understood and that further work might lead to future modifications of the model. For example, the consideration of surface conductance through the partition coefficient might not consider relevant conduction mechanisms, such as the proton hopping or Grotthuss mechanism suggested by Skold *et al.* (2011). Although the pore geometry might not include all characteristics of the material, that are relevant for the temperature dependence, as, for example, the material composition, the comparison of modelling results and experimental findings with regard to the pore size provides a good qualitative agreement

in various respects. However, the model includes many parameters and we have not yet fully exploited the entire parameter space.

5.3 Consequences for practical applications

Seasonal temperature variations mainly affect the first 10 m of the subsurface (e.g. Hayley *et al.* 2007). In the temperate zone, the differences in subsurface temperature are usually in the range of a few Kelvin to 10 K. Considerably larger differences can occur in geothermal areas. As shown by the experiments in the previous sections, temperature variations of a few Kelvin already influence the IP measurement parameters, confirming that temperature dependence might have to be included in the interpretation of IP data. Especially quantitative evaluations based on empirical relationships, which were developed under laboratory conditions, have to be adjusted to different temperatures. A parameter often correlated with different IP parameters and important for geothermal investigations is the hydraulic permeability. For a brief estimation of the influence of temperature we focus on an approach which relates hydraulic permeability k to the time constant. Both quantities depend on a characteristic geometric scale (e.g. Slater 2007), leading to empirical relationships reported by different authors in the form (e.g. Binley *et al.* 2005; Zisser *et al.* 2010b)

$$k = a\tau^b \quad (23)$$

with empirical parameters a and b , which depend on the material. The temperature dependence of the time constant (eq. 6) can easily be inserted into this equation:

$$k(T) = k(T_0) \left[\frac{T_0}{T} \left(\frac{\eta(T)}{\eta(T_0)} \right)^{\alpha_\tau} \right]^b. \quad (24)$$

For Walden exponents $\alpha_{\text{tau}} = 0.5$ –1.4 observed in this work and using $b = 3.85$ as reported by Binley *et al.* (2005) for sandstones, a temperature difference of 10 °C would lead to an error of 43–74 percent in permeability. Therefore, investigations of geothermal fields where the reservoir temperature is considerably higher than the laboratory temperature, ignoring the temperature may lead to a strong underestimation of permeability in the high temperature region.

6 CONCLUSIONS

The investigated samples show a clear temperature dependence of the SIP response, which is apparent both in the real and imaginary part of the conductivity. The absolute maximum of the phase shift appears at different temperatures, depending on the samples. The

characteristic time scales decrease with increasing temperature, displaying different strengths of the temperature dependence (Walden exponents) of different samples. This observation is in good agreement with the decrease with increasing temperature, that is predicted by previous publications. The comparison with CT images indicates that differences in the Walden exponents are related to the pore space geometry. However, we did not quantify robust quantitative relationships with measurable parameters, such as dominant pore-throat diameter or clay content. If we assume that at least 'part of the polarization is caused by an interaction of different pore radii, a correlation with a single parameter characterizing the pore space cannot necessarily be expected.

Furthermore, we simulated the temperature dependence with a membrane polarization model that is based on a sequence of wide and narrow pores. Considering the simplicity of the model geometry, the simulated spectra reproduce several features of the measured data remarkably well. Characteristic times decrease with increasing temperature and the position of the phase maximum is affected by the pore geometry. While the mean pore length mainly controls the spectral position of the maximum phase, the pore radius (or width) determines the temperature, at which the absolute phase maximum occurs. The decrease of the modeled amplitude of the admittance and the characteristic time is stronger than the predicted decrease resulting from ion mobility only. This indicates that, additionally to ion mobility, also other temperature-dependent parameters of the EDL, such as Debye length and Zeta potential, are important.

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APPENDIX A: CONVERSION OF THE 1-D IMPEDANCE TO AN EFFECTIVE CONDUCTIVITY

To convert the 1-D Marshall–Madden impedance (in Ωm^2) to an effective resistivity, we consider a unit cell consisting of a pore sequence (one wide and one narrow cylindrical pore) within a cuboid of length $L = L_1 + L_2$ and a cross-sectional area $A_S = a_S^2$ with $a_S \geq 2 \cdot r_1$. The volume surrounding the pores is assumed to be non-conducting. The porosity of this unit cell is given by

$$\Phi = \frac{A_1 L_1 + A_2 L_2}{A_S L}. \quad (\text{A1})$$

By dividing the Marshall–Madden impedance by the cross-sectional area of the larger pore $Z = Z_{\text{MM}}/A_1$, it is converted to the resistance of the pore sequence. From these considerations, we get the effective resistivity of the unit cell

$$\rho_{\text{eff}} = Z \cdot \frac{A_S}{L} \quad (\text{A2})$$

$$= \frac{Z_{\text{MM}}}{A_1} \cdot \frac{A_1 L_1 + A_2 L_2}{L^2 \Phi}, \quad (\text{A3})$$

and the effective conductivity $\sigma_{\text{eff}} = 1/\rho_{\text{eff}}$.

APPENDIX B: CONSIDERATION OF A REDUCED ION MOBILITY IN THE STERN LAYER

A reduced ion mobility in the Stern layer μ_s can be included to the membrane polarization model by modifying the equations of the Stern layer correction in Bückner & Hördt (2013a):

$$2\pi r \left(\sigma_d + \frac{\mu_s}{\mu} \sigma_s \right) = F c_0 A_p (\tilde{b}_p - 1) \quad (\text{B1})$$

with the surface charge densities associated with the Stern and the diffuse layer σ_d and σ_s and the pore section A_p . Following Bückner & Hördt (2013a), we get for the average cation concentration

$$\tilde{b}_p = \frac{\bar{b}_p \left[1 + f_Q \left(\frac{\mu_s}{\mu} - 1 \right) \right] - f_Q \frac{\mu_s}{\mu}}{1 - f_Q}, \quad (\text{B2})$$

which in the case $\mu_s = \mu$ becomes eq. (16). In the case $\mu_s \ll \mu$ eq. (B2) reduces to $\tilde{b}_p \approx \bar{b}_p$.